

Experiment 'p_2_10_hybrid_car_cost' Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_2_10_hybrid_car_cost

Natural language statement: null

Method signature: p_2_10_hybrid_car_cost(new_car_cost: real, miles_per_year: real, gas_price: real, mpg: real, resale_value: real) returns (total_cost: real)

Ensures

- `total_cost == new_car_cost + (miles_per_year / mpg) * gas_price * 5.0 - resale_value`

Requires

- `0.0 < mpg`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

```
p_2_10_hybrid_car_cost(new_car_cost: real, miles_per_year: real, gas_price: real, mpg:
    ↪ real, resale_value: real) returns (total_cost: real)
```

The method should respect the following contract:

```
requires 0.0 < mpg, ensures total_cost == new_car_cost + (miles_per_year / mpg) *
    ↪ gas_price * 5.0 - resale_value
```

Produce and show only the Dafny body of this method, including the curly braces that
↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
↪ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn30oCDZIJw64wgui48wUtd2XIPoy

Model: gpt-5.5-2026-04-23

Created at: 1780582022

Finish reason: stop

Usage: 350 tokens in, and 83 tokens out

```
{
  total_cost := new_car_cost + (miles_per_year / mpg) * gas_price * 5.0 - resale_value;
}
```

Final Program

```
method p_2_10_hybrid_car_cost(new_car_cost: real, miles_per_year: real, gas_price: real,
    ↪ mpg: real, resale_value: real) returns (total_cost: real)
  requires 0.0 < mpg
  ensures total_cost == new_car_cost + (miles_per_year / mpg) * gas_price * 5.0 -
    ↪ resale_value
{
  total_cost := new_car_cost + (miles_per_year / mpg) * gas_price * 5.0 - resale_value;
}
```

Total Token Usage

Input tokens: 350

Output tokens: 83

Reasoning tokens: 34

Sum of 'total tokens': 433

Experiment Timings

Overall Experiment started at 1780582021678, ended at 1780582058205, lasting 36527ms (36.53 seconds)

Iteration #1 started at 1780582021679, ended at 1780582058205, lasting 36526ms (36.53 seconds)